



National Institute of Technology Rourkela
Department of Computer Science Engineering
Deep Learning (CS6510)
Mid-Semester Examination
B.Tech 4th year/PhD/M.Tech(R)

Duration: 2 hrs

Spring 2026, 8th Semester

Total marks: 30

Attempt all questions;

⟨CO-x: Course Outcome level⟩; ⟨BL-x: Bloom's taxonomy level⟩;

**All parts of the same question must be answered together*

1. (a) State the Perceptron Learning algorithm (PLA). If the sets P (positive samples) and N (negative samples) are finite and linearly separable, prove that the PLA updates the weight vector W a finite number of times such that the two sets are separated. ⟨CO-1,2⟩ ⟨BL-2, 3⟩ [5]
(b) Using a single-layer perceptron, simulate the following combinatorial circuit elements. Find the Weight vectors (W_i) and Bias (β) for each of the following. The circuit has two inputs x_1, x_2 . ⟨CO-2⟩ ⟨BL-2, 3⟩ [5]
 - i) NOR gate
 - ii) NAND gate
 - iii) AND gate followed by an OR gate with the second gate (OR) receiving two inputs: first as the output from the AND gate, and second from the source input x_2 .
2. (a) Define Gradient Descent (GD) in the context of training an ANN. Compare and contrast all the variants of GD with appropriate mathematical formulations. ⟨CO-3⟩ ⟨BL-3⟩ [5]
(b) Suppose a single-layer perceptron with two inputs and one output neuron uses the binary sigmoid activation function. Given the parameters $\bar{w} = [0.2 \ -0.3]^\top$, $b = 0.1$, and learning rate $\eta = 0.01$, determine the neuron output for each input pattern and update $\bar{w}$ and b for two epochs. Use mean squared error loss and the steepest (gradient) descent method. ⟨CO-4,5⟩ ⟨BL-5⟩ [5]

Example	x_1	x_2	Class
1	2.3	1.5	0
2	0.7	3.1	0
3	4.2	2.8	1
4	5.0	4.1	1

3. (a) What are the disadvantages of using a fixed learning rate η ? Compare and contrast various optimizers for tuning the time-dependent learning rate η_t with appropriate mathematical formulations. ⟨CO-3⟩ ⟨BL-3⟩ [3]
(b) Consider the following ANN architecture with dataset $\mathcal{R}^{d=3}$:
 - Input layer (IP): d neurons

- First hidden layer (H_1): 3 neurons
- Second hidden layer (H_2): 3 neurons
- Output layer (OP): 1 neuron
- Activation function: Sigmoid for all the layers

$$X = [2, 3, 5]^T$$

$$W_{IP-H_1} = [1.2, 2.3, -0.9, 2.3, 3.5, -7.8, 8.7, -9.1, 4.3]^T$$

$$W_{H_1-H_2} = [1.2, 2.5, -0.1, -2.3, 3.5, -5.8, 8.7, -9.1, 1.3]^T$$

$$W_{H_2-OP} = [-1.2, 4.5, -5.8]^T$$

$$\beta_{H_1} = 0.8 \text{ (all the neurons)}$$

$$\beta_{H_2} = 0.3 \text{ (all the neurons)}$$

$$\beta_{OP} = 0.2 \text{ (all the neurons)}$$

Perform two epochs over this ANN to learn the updated weights and bias. Find whether the above ANN suffers from Vanishing Gradient (VGD)? Compute the fraction of neurons that needs to be turned off if there is a VGD. The range of allowed update for VGD is $\leq 10^{-4}$ ⟨CO-3⟩ ⟨BL-3⟩ [7]

N.B: Make your own logical assumptions (if required) to answer wherever necessary.